Velocity-Dependent Mass Functional, Preferred-Frame Effects, and Intrinsic Temporal Stochasticity

Omar Iskandarani[*]

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Abstract

We analyze the weak-field experimental consequences of Swirl-String Theory (SST) in its covariant vector–tensor formulation. Building on the master mass functional derived in Atomic Masses from Topological Invariants of Knotted Field Configurations and the clock–foliation structure fixed in Canon v0.7.7, we show that preferred-frame effects arise as velocity-dependent corrections to the inertial and gravitational mass functional, rather than from additional forces or modified gravitational propagation. Imposing the observational constraint from GW170817 that gravitational waves propagate at the speed of light reduces the accessible parameter space to a narrow sector governed by the combinations c_{14} and c_2. We propose a tabletop null experiment based on a rotating cryogenic sapphire resonator system to provide an independent laboratory constraint on the Parameterized Post-Newtonian preferred-frame parameters α_1 and α_2.

[*] **Independent Researcher, Groningen, The Netherlands**
Email: info@omariskandarani.com
ORCID: 0009-0006-1686-3961
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General Relativity describes gravitation as spacetime geometry while treating inertial mass and physical time as primitive inputs. Swirl–String Theory (SST) departs from this paradigm by deriving both mass and time from a single covariant action involving a Lorentzian metric and a unit timelike vector field.

SST-59 established a master mass functional based on the integrated swirl energy density. Canon v0.7.7 unified this construction with a clock–foliation field u^μ, equivalent in the infrared to the Einstein–Æther or khronometric framework. The remaining open question is how motion relative to this foliation manifests operationally and experimentally.

The present work should therefore be read as an operational and experimental follow-up to the master mass functional derived in SST-59, with no modification of its foundational assumptions. Specifically, this paper provides two results: (i) a canonical identification of preferred-frame effects with velocity-dependent corrections to the mass functional, and (ii) an operational framework for probing intrinsic temporal stochasticity. These two results are logically independent but conceptually linked. Velocity-dependent mass corrections probe anisotropy of the gravitational sector, while intrinsic temporal stochasticity probes the operational meaning of time itself. The latter is developed formally in Sect. 5 and supported by classical and quantum analogues discussed throughout the appendices.

1 Mass Functional and Clock Foliation

In SST, inertial and gravitational mass are defined by

$$M = \mathcal{C} \int_V \rho_E(\mathbf{x})\, dV, \tag{1}$$

where ρ_E is the local swirl energy density and $\mathcal{C}$ is a fixed normalization constant [20].

The evaluation of ρ_E is referenced to a local clock–foliation structure represented covariantly by a unit timelike vector field

$$u^\mu u_\mu = -1. \tag{2}$$

2 Velocity-Dependent Mass Functional

A system moving with spatial velocity $\mathbf{v}$ relative to the foliation samples a boosted slicing of the clock field. The effective mass therefore becomes velocity dependent:

$$M_{\text{eff}}(\mathbf{v}) = M_0 \left[1 + \alpha_1 \frac{\mathbf{v}\cdot\hat{u}}{c} + \alpha_2 \frac{(\mathbf{v}\cdot\hat{u})^2}{c^2} + \mathcal{O}\left(\frac{v^3}{c^3}\right) \right], \tag{3}$$

where $\hat{u}$ denotes the spatial direction selected by the clock field. The coefficients α_1 and α_2 quantify anisotropic corrections to the mass functional rather than new interactions.

[Electromagnetic Non-Generation of Gravity]

In Swirl–String Theory (SST), electromagnetic fields *cannot* directly modify gravitational mass, inertial response, or spacetime curvature. Any apparent electromagnetic influence on gravity must proceed *indirectly* through the generation of a persistent foliation pressure in the underlying medium.

More precisely, let χ denote the foliation (clock) field and ρ_E the swirl energy density. A necessary and sufficient condition for an electromagnetic configuration to induce a gravitational effect is the existence of a non-radiative, long-lived contribution

$$\Delta mc^2 \;=\; \int_V \Delta p_{\text{eff}}\, dV \;=\; \int_V \Delta\rho_E\, dV, \tag{4}$$

such that

$$\Delta(\nabla_\mu \chi) \neq 0 \quad \text{after all electromagnetic fields are removed.} \tag{5}$$

Electromagnetic energy that is oscillatory, radiative, gauge-removable, or time-averaged to zero cannot satisfy this condition and therefore cannot generate or modify gravity in SST.

This theorem fixes the admissible sources of the SST mass functional and excludes electromagnetic configurations as primary generators of gravitational structure, independent of gauge choice or time averaging.

3 Mapping to Einstein–Æther and PPN Formalism

The covariant SST action shares its infrared structure with Einstein–Æther theory. Imposing the observational constraint from GW170817 that the speed of gravitational waves equals the speed of light requires

$$c_{13} = c_1 + c_3 = 0. \tag{6}$$

Under this constraint, the preferred-frame PPN parameters reduce to

$$\alpha_1^{\mathrm{SST}} = -4c_{14}, \qquad \alpha_2^{\mathrm{SST}} = \alpha_2(c_{14}, c_2), \tag{7}$$

where $c_{14} = c_1 + c_4$ and c_2 governs the scalar sector [21].

3.1 Relation to Wave-Optical Gravitational Lensing

Recent work on gravitational-wave lensing in the wave-optics regime has clarified that interference phenomena near caustics are governed primarily by phase geometry rather than by amplitude magnification alone. In particular, Bulashenko and Ubach have shown that the transition between diffraction, amplification, and geometrical-optics oscillations is controlled by a two-parameter criterion involving both the Fresnel number and the angular source displacement relative to the Einstein angle, rather than by the wavelength-to-Schwarzschild-radius ratio alone [7]. Close to a caustic, geometrical optics breaks down at any finite frequency, and the resulting beating patterns exhibit universal nodal and antinodal structures corresponding to constant phase differences between multiple propagation paths. A fixed $\pi/2$ phase offset associated with saddle points of the time-delay function appears as a topological (Morse) contribution, independent of the detailed lens model. These results provide a concrete demonstration that caustics represent degeneracies in the effective time-of-flight structure, where multiple histories contribute coherently to the observed signal. From the perspective of Swirl-String Theory, such behavior admits a natural interpretation in terms of foliation degeneracies of the swirl-clock structure, in which distinct flow-guided trajectories share a common projected time coordinate. While the present analysis does not alter the predictions of general relativity, it establishes a robust observational regime in which phase-topological effects dominate, thereby defining a clean baseline against which SST-specific deviations may be formulated and tested.

4 Experiment A: Gravity-Sector Preferred-Frame Null Test

4.1 Physical Principle

Because SST preferred-frame effects modify the inertial mass functional, any resonant system whose eigenfrequencies depend on effective mass becomes a probe of gravity- sector anisotropy, independent of electromagnetic Lorentz violation.

4.2 Apparatus

The proposed apparatus consists of two orthogonal cryogenic sapphire resonators mounted in a single vacuum cryostat at 4.2 K and rotated on a precision air-bearing turntable with angular

frequency $\omega_{\rm rot}$. The differential resonance frequency

$$\Delta f(t) = f_x(t) - f_y(t) \tag{8}$$

is monitored using heterodyne readout with fractional stability $\Delta f/f \sim 10^{-17}$.

4.3 Signal Model

The expected signal admits the harmonic expansion

$$\frac{\Delta f}{f} = S_2 \cos(2\omega_{\rm rot} t) + \sum_{n=\pm1,\pm2} S_{2,n} \cos(2\omega_{\rm rot} t + n\omega_\oplus t), \tag{9}$$

where $\omega_\oplus$ is the sidereal frequency. Sidereal sidebands uniquely identify preferred-frame effects.

Phase–topological origin of intrinsic temporal fluctuations. Experimental and theoretical studies of vortexed photon states demonstrate that geometric and topological phase structures can induce fixed phase offsets and interference patterns that are insensitive to local field amplitudes and device details. In particular, Pancharatnam–Berry–Gouy phase effects introduce discrete phase discontinuities across focusing and mode–conversion processes, reflecting degeneracies in the effective time–of–flight structure where multiple propagation histories contribute coherently. Such degeneracies naturally lead to observable phase jitter and mode–dependent timing fluctuations even in otherwise deterministic classical systems. While unrelated to gravitation, these results provide an explicit laboratory realization of how intrinsic temporal broadening may arise from phase–topological structure alone. In this sense, the intrinsic temporal stochasticity introduced in Sect. 5 can be interpreted as a generic consequence of foliation–dependent phase organization, rather than as a phenomenological noise source.

5 Intrinsic Temporal Stochasticity

In SST-31, time is defined as a relational observable associated with a conserved event current. This implies the possibility of intrinsic fluctuations in clock readouts.

The observed time-of-arrival distribution is modeled as

$$P_{\rm obs}(\Theta) = \int dt\, P_{\rm cl}(t) \frac{1}{\sqrt{2\pi\sigma_\tau^2}} \exp\left[-\frac{(\Theta - t)^2}{2\sigma_\tau^2}\right], \tag{10}$$

where σ_τ^2 characterizes intrinsic clock noise. The Canon implies that σ_τ^2 may depend on the gradient of the clock potential,

$$\sigma_\tau^2 = \sigma_\tau^2(\nabla\chi), \tag{11}$$

with its detailed spectrum left phenomenological.

Experimental studies of single–bubble sonoluminescence demonstrate that energy emission exhibits large, reproducible timing variations uncorrelated with peak compression or temperature, indicating that emission is governed by internal phase organization rather than local energetic extrema [18].

Foliation-dependent dynamics in extreme hydrodynamic collapse. Independent support for foliation- and phase-dependent dynamics, conceptually analogous to the intrinsic temporal stochasticity defined in Sect. 5, is provided by refined hydrodynamic studies of single–bubble sonoluminescence. Detailed numerical simulations demonstrate that light emission during bubble collapse is not determined by peak temperature or energy density alone, but depends sensitively

on the internal spatiotemporal structure of the flow, including vapor content, chemical reactions, and the emergence (or suppression) of coherent compressive cores. In particular, identical external driving conditions may lead either to shock–like behavior or to smooth compression waves depending on the internal configuration, indicating that regime transitions are controlled by structural rather than purely energetic criteria. These results show that extreme emission phenomena can arise from transient degeneracies in the effective time and phase organization of the system, even in a purely classical fluid setting. While unrelated to gravitation, such behavior supports the broader SST perspective that physically observable effects may be governed by foliation–dependent dynamics rather than by local field amplitudes alone.

Chirality, orientation, and matter–antimatter asymmetry. Classical studies of tight composite knots demonstrate that chirality constitutes an intrinsically energy–bearing geometric property, rather than a purely topological label. In particular, chiral and achiral composites constructed from the same prime knots exhibit systematic differences in elastic energy in their tight configuration, with the dominant contribution arising from internal torsion. Since torsion is sensitive to orientation, these results establish that opposite geometric handedness corresponds to physically inequivalent energetic states, even in the absence of microscopic interactions or external fields. While derived within classical rope elasticity and unrelated to particle physics, this finding provides a structurally relevant precedent for theories in which matter and antimatter are distinguished by orientation or internal rotational structure rather than by independent dynamical degrees of freedom. In the SST framework, this supports the interpretation that matter–antimatter distinction may be encoded at the level of geometric orientation of otherwise equivalent underlying configurations.[2] Further structural support for intrinsic temporal stochasticity is provided by: Appendix A (relational phase evolution), Appendix D (compact time and bounded evolution), and Appendix G (holonomy of the swirl–clock phase).

6 Experiment B: Exploratory Clock-Noise Search

Experiment B operationalizes the intrinsic temporal stochasticity introduced in Sect. 5. It employs a pair of co-located optical lattice clocks operated in differential mode to suppress common-mode noise. The observable is an anomalous decoherence rate

$$\Gamma_{\rm obs} = \Gamma_{\rm env} + \Gamma_{\rm SST}, \qquad \Gamma_{\rm SST} \propto \sigma_\tau^2(\nabla\chi), \tag{12}$$

correlated with controlled modulation of the local clock-field gradient. This experiment defines a search channel rather than a guaranteed detection.

7 Discussion and Conclusion

The analysis presented here demonstrates that Swirl–String Theory predicts two distinct but experimentally accessible consequences: anisotropic corrections to the mass functional associated with preferred foliation, and intrinsic temporal stochasticity arising from relational time. Neither effect requires modifications of gravitational wave propagation, electromagnetic dynamics, or Lorentz invariance in the infrared.

Together, the proposed experiments define a minimal and falsifiable low-energy test program for SST consistent with existing gravitational constraints.

Recent potential-based analyses show that quark mixing angles and the weak CP-violating phase arise as extrema of flavor-invariant mass functionals, rather than as independent input parameters [22]. Within Swirl–String Theory, this behavior is naturally interpreted as the variational alignment of closed configurations that minimize a geometric energy functional built from invariant combinations of masses. CP violation then emerges as a nontrivial phase at the minimum of this functional, consistent with its interpretation as a holonomy of relational time.

A Emergence of the Schrödinger Equation from Relational Time

This appendix provides a formal derivation supporting the operational notion of relational time used in Sect. 5. It does not introduce new physical assumptions beyond those already employed in the main text.

A.1 Relational Time and Phase Dynamics

In Swirl–String Theory (SST), physical time is not introduced as a fundamental external parameter, but arises operationally from a preferred foliation defined by a unit timelike vector field u^μ. Observables evolve along the integral curves of u^μ, and the physically meaningful time derivative is the directional derivative

$$\frac{D}{Dt} \equiv u^\mu \nabla_\mu. \tag{13}$$

This construction parallels the "khronon" formulation of hypersurface–orthogonal Einstein–Æther theory and provides a covariant notion of simultaneity.

Consider a complex scalar field $\Psi(x)$ representing a localized excitation of the SST medium. The field is assumed to admit a rapidly oscillating internal phase associated with a characteristic rest energy $E_0 = mc^2$, where the effective mass m is determined by the SST mass functional (see SST–59). We therefore write

$$\Psi(\mathbf{x}, t) = \psi(\mathbf{x}, t) \exp\left(-\frac{i}{\hbar} mc^2 t\right), \tag{14}$$

where ψ is a slowly varying envelope with respect to the clock time defined by the foliation.

A.2 Underlying Relativistic Dynamics

At wavelengths large compared to the SST core scale r_c, the excitation dynamics are governed by a second–order hyperbolic equation consistent with covariance and locality. To leading order, this takes the Klein–Gordon form

$$\left(\frac{1}{c^2}\frac{D^2}{Dt^2} - \nabla_\perp^2 + \frac{m^2c^2}{\hbar^2}\right)\Psi = 0, \tag{15}$$

where $\nabla_\perp^2$ is the Laplacian on spatial hypersurfaces orthogonal to u^μ. Equation (15) does not introduce new degrees of freedom beyond those already present in the SST covariant action; it represents the infrared wave dynamics of a massive excitation propagating along the foliation.

A.3 Nonrelativistic Limit

Substituting the factorization (14) into (15) and assuming the nonrelativistic regime

$$\left|\frac{D\psi}{Dt}\right| \ll \frac{mc^2}{\hbar}|\psi|, \qquad |\nabla_\perp \psi| \ll \frac{mc}{\hbar}|\psi|, \tag{16}$$

one finds that second–order time derivatives of ψ are parametrically suppressed. Retaining terms to leading order in v/c, the envelope equation reduces to

$$i\hbar\frac{D\psi}{Dt} = -\frac{\hbar^2}{2m}\nabla_\perp^2\psi. \tag{17}$$

Equation (17) is precisely the free Schrödinger equation, with the time derivative replaced by evolution along the SST clock foliation.

A.4 External Potentials and Clock Gradients

Weak interactions with background fields or inhomogeneities of the SST medium enter as perturbations to the local clock rate. A scalar clock potential Φ_χ modifies the rest energy as

$$mc^2 \;\to\; mc^2 + V, \qquad V \equiv m\Phi_\chi, \tag{18}$$

leading to the full nonrelativistic equation

$$i\hbar \frac{D\psi}{Dt} = \left(-\frac{\hbar^2}{2m}\,\nabla_\perp^2 + V \right) \psi. \tag{19}$$

In SST, the potential V is interpreted as a pressure or clock–rate perturbation associated with gradients of the foliation field, rather than as a fundamental force.

A.5 Interpretation

Equation (19) demonstrates that the Schrödinger equation arises in SST as a controlled infrared limit of relativistic phase evolution along a preferred foliation. The mass parameter m is not fundamental, but encodes stored energy of the underlying medium through the SST mass functional. The complex wavefunction ψ represents the slowly varying envelope of a high–frequency internal clock.

In this sense, nonrelativistic quantum mechanics is not postulated but emerges as the effective phase–transport equation governing relational time evolution in the SST framework.

A.6 Domain of Validity

The derivation holds under the following conditions:

- velocities small compared to c,
- wavelengths large compared to the SST core scale r_c,
- weak clock–rate gradients $\nabla\Phi_\chi$.

Corrections to Schrödinger dynamics are expected at higher order in v/c and from stochastic clock fluctuations, discussed separately in the context of intrinsic time broadening.

A.6.1 Normalizability and Finite Clock Evolution

The derivation of the Schrödinger equation in Section A implicitly assumes that the clock parameter governing phase evolution defines a normalizable quantum state. If the clock parameter were unbounded, wave packets evolving along the foliation would generically fail to remain square–integrable.

Independent analyses of quantum mechanics in accelerated and gravitational frames demonstrate that unbounded time evolution leads to pathological Hilbert spaces unless the associated boost or clock parameter is restricted to a finite domain [10, 6].

In SST, this restriction is implemented geometrically by the compactness of the swirl–clock variable,

$$S^{\circlearrowleft}_{(t)} \in \mathbb{S}^1, \tag{20}$$

which ensures that Schrödinger evolution remains well defined and that quantum states are normalizable at all times. The emergence of nonrelativistic quantum mechanics is therefore inseparable from the bounded structure of relational time.

B Discrete Topological Origin of Fermionic Mass Hierarchies

B.1 Motivation

In the Standard Model, fermion masses and mixing angles are encoded in arbitrary Yukawa matrices. In Swirl–String Theory (SST), by contrast, mass emerges from geometric and topological properties of closed swirl–string configurations embedded in a preferred foliation.

It has been shown that hierarchical fermion mass spectra may arise from purely discrete order counting without invoking continuous flavor symmetries [11, 12]. Here we demonstrate that SST provides a deeper physical origin for such order structures by identifying them with intrinsic topological suppression factors of swirl–string configurations.

B.2 Discrete Geometric Suppression Parameter

Consider a closed swirl–string configuration K characterized by:

- core radius r_c,
- effective string length L_K,
- topological complexity index $n_K \in \mathbb{Z}_{\geq 0}$,

where n_K encodes knotting, linking, and internal winding complexity.

We define the dimensionless geometric suppression parameter

$$\epsilon_K \equiv \frac{r_c}{L_K}, \qquad \epsilon_K \ll 1. \tag{21}$$

This parameter is fixed by geometry and is not a free coupling.

B.3 Mass Functional as a Topological Power Series

The SST mass functional associated with configuration K may be written as

$$M_K = \mathcal{A}\, \rho_E\, V_K\, \epsilon_K^{n_K}, \tag{22}$$

where:

- ρ_E is the swirl energy density,
- V_K is the effective core volume,
- n_K is the topological order index,
- $\mathcal{A}$ is a universal normalization fixed by the SST mass scale.

Equation (22) shows that fermion mass hierarchies arise as integer powers of a single geometric ratio, rather than from arbitrary Yukawa coefficients.

B.4 Generation Structure from Topological Order

Configurations of lowest complexity ($n_K = 0$) acquire unsuppressed masses, while higher values of n_K produce progressively lighter fermions. This yields the qualitative hierarchy

$$M^{(3)} \gg M^{(2)} \gg M^{(1)}, \qquad \frac{M^{(i)}}{M^{(j)}} \sim \epsilon_K^{n_i - n_j}. \tag{23}$$

The heaviest fermion corresponds to a minimal, topologically protected swirl–string configuration, while lighter fermions correspond to configurations with additional internal winding that dilute swirl energy over larger effective lengths.

B.5 Discrete Phase, Mixing, and CP Violation

Closed swirl–strings admit discrete circulation phases compatible with foliation closure. We associate a phase label

$$\theta_K = \frac{2\pi k}{N}, \qquad k \in \mathbb{Z}_N, \tag{24}$$

with each configuration.

Observable mixing arises from geometric misalignment between swirl–string embeddings in different sectors, while CP violation corresponds to a nontrivial relative swirl phase between otherwise real mass functionals. No fundamental complex Yukawa couplings are required.

B.6 Neutrino Sector as Nested Swirl Excitations

Light neutrino masses arise naturally as second–order excitations of nested swirl–string structures. Small loops couple indirectly to larger, stiffer configurations, producing effective masses of the form

$$M_\nu \;\sim\; \epsilon_K^2\, M_{\text{swirl}}, \tag{25}$$

analogous in structure to seesaw suppression but originating purely from geometry [13, 14].

B.7 Experimental Implications

Flavor–changing effects correspond to partial geometric overlap of distinct swirl–string configurations. Neutral meson mixing bounds therefore constrain allowable inter–configuration coupling strengths, while preferred–frame and mass–functional anisotropy tests probe the same underlying structure.

B.8 Summary

Fermion mass hierarchies in SST emerge from integer–ordered topological suppression, require no continuous flavor symmetry, and unify mass generation, mixing, and CP violation as consequences of swirl geometry and foliation structure.

C Mass–Ratio Origin of Quark Mixing

An explicit construction exists in which the Cabibbo–Kobayashi–Maskawa (CKM) matrix can be derived directly from quark mass ratios, without introducing Yukawa textures or flavor symmetries [8].

In this approach, an effective geometric radius R_i is associated with each quark q_i through a cubic scaling of its mass,

$$R_i \;\propto\; M_i^{1/3}. \tag{26}$$

The interaction strength between two quark flavors is then assumed to depend on the ratio of these effective radii,

$$Q_{ij} \;=\; \frac{R_i}{R_j} \;=\; \left(\frac{M_i}{M_j}\right)^{1/3}. \tag{27}$$

A unitary quark mixing matrix is subsequently constructed from Q_{ij} via a matrix exponential,

$$V_Q \;=\; \exp\!\left[-i\,\Delta\,(S + S^T)\right], \tag{28}$$

where S is a matrix formed from Q_{ij} and $\Delta = \det Q$. Remarkably, this procedure reproduces the observed hierarchical structure of the CKM matrix using only quark masses as input.

C.1 Interpretation within Swirl–String Theory

Within SST, fermion masses arise from the energy content of closed swirl–string configurations,

$$M_K \;\propto\; \rho_E\, V_K, \tag{29}$$

where V_K is the effective core volume of configuration K. Identifying $V_K \sim L_K^3$, where L_K is the characteristic swirl–string length, equation (27) acquires a direct geometric meaning,

$$\left(\frac{M_i}{M_j}\right)^{1/3} \;\longleftrightarrow\; \frac{L_i}{L_j}. \tag{30}$$

Quark mixing in SST is therefore interpreted as arising from the geometric overlap and relative embedding of swirl–string configurations with different characteristic lengths. The observed suppression of off–diagonal CKM elements follows naturally from large disparities in swirl–string scales.

C.2 Conceptual Consequences

This construction supports the SST claim that:

- mixing angles are not fundamental parameters,
- flavor mixing is a derived geometric effect,
- mass hierarchies and mixing hierarchies share a common origin.

In SST, the matrix exponential form reflects continuous deformations of swirl–string embeddings rather than ad hoc unitary rotations in flavor space.

D Energy Saturation from Quantum Gravitational Foliation

D.1 Motivation

Recent work by Mäkelä has shown that quantum gravitational consistency alone, without Lorentz violation or modified dispersion relations, enforces a finite upper bound on the attainable energy of a massive particle [6]. The bound arises from the requirement that quantum states perceived by uniformly accelerated observers remain normalizable within a finite boost domain.

This result is directly relevant to Swirl–String Theory (SST), where mass and time are defined relative to a compact clock–foliation structure and where the mass functional is finite by construction.

D.2 Quantum–Mechanical Consistency and Time Compactness

Beyond numerical considerations, the existence of a maximum attainable energy has a deeper conceptual origin. Recent analyses of quantum mechanics in gravitational and accelerated frames show that the standard formulation of quantum theory becomes inconsistent if the time parameter is assumed to be globally unbounded [10].

In particular, when quantum states are defined relative to observers with constant proper acceleration, the requirement of Hilbert–space normalizability enforces a finite domain for the associated time or boost parameter. Extending time evolution beyond this domain leads to non–physical, non–normalizable states.

This result implies that compact or effectively bounded time is not an additional assumption, but a necessary condition for the existence of a well–defined quantum theory in the presence of gravitation. In Swirl–String Theory (SST), this requirement is naturally satisfied by the compact swirl–clock variable $S^{\circlearrowleft}_{(t)}$, whose bounded domain guarantees quantum consistency and directly yields the observed energy saturation.

D.3 Quantum Gravitational Energy Bound

For a particle of rest mass m, Mäkelä derives the universal bound

$$E_{\max} = mc^2 \cosh\left(\frac{2\pi^2}{\ln 2}\right), \tag{31}$$

independent of the particle species or interaction details.

Numerically,

$$\cosh\left(\frac{2\pi^2}{\ln 2}\right) \simeq 1.17 \times 10^{12}, \tag{32}$$

$$E_{\max}^{(p)} \simeq 1.1 \times 10^{21}\,\mathrm{eV}, \tag{33}$$

$$E_{\max}^{(e)} \simeq 6 \times 10^{17}\,\mathrm{eV}, \tag{34}$$

placing the effect far below the Planck scale yet within reach of ultra–high–energy cosmic ray observations.

D.3.1 Numerical evaluation

Mäkelä's bound may be written in terms of a maximum rapidity

$$\phi_{\max} = \frac{2\pi^2}{\ln 2}, \qquad \gamma_{\max} = \cosh(\phi_{\max}), \qquad \beta_{\max} = \tanh(\phi_{\max}), \tag{35}$$

so that $E_{\max} = \gamma_{\max}\, mc^2$ [6].

Evaluating the universal factors gives

$$\phi_{\max} \simeq 28.47765865, \tag{36}$$

$$\gamma_{\max} \simeq 1.165896540231 \times 10^{12}, \tag{37}$$

$$1 - \beta_{\max} \simeq \frac{1}{2\gamma_{\max}^2} \simeq 3.67832397 \times 10^{-25}. \tag{38}$$

Hence $\beta_{\max}$ is extremely close to unity, and the ceiling is dominated by $\gamma_{\max}$.

Using standard rest energies as inputs, $m_p c^2 \simeq 0.938\,272\,081\,3\,\mathrm{GeV}$ and $m_e c^2 \simeq 0.510\,998\,95\,\mathrm{MeV}$, one obtains

$$E_{\max}^{(p)} = \gamma_{\max} m_p c^2 \simeq 1.0939 \times 10^{21}\,\mathrm{eV} \simeq 1.7527 \times 10^{2}\,\mathrm{J}, \tag{39}$$

$$E_{\max}^{(e)} = \gamma_{\max} m_e c^2 \simeq 5.9577 \times 10^{17}\,\mathrm{eV} \simeq 9.545 \times 10^{-2}\,\mathrm{J}, \tag{40}$$

where we used $1\ \mathrm{eV} = 1.602\,176\,634 \times 10^{-19}\,\mathrm{J}$ [9].

In SST notation, the same numerical ceiling applies multiplicatively to the rest energy functional

$$E_0(K) = \mathcal{C}\, c^2 \int_{V_K} \rho_E(\mathbf{x})\, dV, \tag{41}$$

so that

$$E_{\max}(K) = \gamma_{\max}\, E_0(K), \qquad \gamma_{\max} = \cosh\left(\frac{2\pi^2}{\ln 2}\right). \tag{42}$$

All model dependence remains confined to $E_0(K)$; the factor $\gamma_{\max}$ is universal and fixed by the quantum–gravitational normalizability condition [6].

D.4 Reduction to the SST Mass Functional

In SST, inertial and gravitational mass are not fundamental parameters but are defined by the master mass functional

$$M = \mathcal{C} \int_V \rho_E(\mathbf{x})\, dV, \tag{43}$$

where ρ_E is the local swirl energy density evaluated relative to the clock–foliation field.

The corresponding rest energy of an excitation K is therefore

$$E_0(K) = Mc^2 = \mathcal{C}\, c^2 \int_{V_K} \rho_E(\mathbf{x})\, dV. \tag{44}$$

Substituting equation (44) into the quantum gravitational bound (31), we obtain the SST form of the energy ceiling:

$$\boxed{E_{\max}(K) = \mathcal{C}\, c^2 \cosh\left(\frac{2\pi^2}{\ln 2}\right) \int_{V_K} \rho_E(\mathbf{x})\, dV}. \tag{45}$$

The factor $\cosh(2\pi^2/\ln 2)$ is universal and kinematic; all model dependence enters solely through the SST mass functional.

D.5 Interpretation in Terms of Clock Foliation

The hyperbolic cosine in equation (45) reflects a finite range of admissible boost angles for normalizable quantum states. In SST language, this corresponds to the compactness of the swirl–clock phase,

$$S^{\circlearrowleft}_{(t)} \in \mathbb{S}^1. \tag{46}$$

Allowing unbounded evolution along the clock direction would render the quantum state non–normalizable, precisely as in the gravitational derivation of [6]. Energy saturation is therefore a kinematic consequence of compact foliation time, not a dynamical cutoff.

[Quantum–Gravitational Time Boundedness]

Any quantum theory admitting a gravitationally consistent observer description requires that the physical time parameter governing state evolution be compact or effectively bounded.

Sketch. For observers undergoing constant proper acceleration, quantum states are naturally described in terms of boost–adapted time coordinates. If the associated time or rapidity parameter is unbounded, the resulting quantum states fail to be normalizable in Hilbert space. This pathology arises independently of Lorentz violation or ultraviolet modifications and is a direct consequence of combining quantum mechanics with general relativity [10, 6].

Therefore, a finite domain of physical time is required for quantum consistency. In Swirl–String Theory, this condition is satisfied by the compact swirl–clock phase $S^{\circlearrowleft}_{(t)}$, which enforces both quantum normalizability and the existence of a maximum attainable energy. □

D.6 Physical Consequences

Equation (45) implies that:

- the maximum attainable energy of a massive excitation is finite,
- the bound scales linearly with the SST mass functional,
- no Planck–scale physics is required,
- ultra–high–energy cosmic rays probe foliation geometry rather than acceleration mechanisms.

Thus, SST does not introduce an ad hoc energy cutoff; it explains why such a cutoff is required by quantum gravitational consistency.

D.7 Summary

The quantum gravitational energy bound derived by Mäkelä reduces in SST to a universal multiplicative factor acting on the master mass functional. Energy saturation is therefore an unavoidable consequence of relational time and compact clock foliation, fully consistent with the SST framework.

E Modular Symmetry as Emergent Swirl–Clock Geometry

E.1 Motivation

Recent developments in flavor physics have demonstrated that lepton masses and mixing angles can be reproduced using modular invariance, where all flavor structure is encoded in modular forms of a single complex parameter τ [15, 16]. In such constructions, τ acts as a spurionic modulus whose vacuum value determines mass hierarchies, mixing angles, and CP–violating phases.

While phenomenologically successful, these models leave unanswered the fundamental physical origin of τ and the reason modular functions should govern fermion properties. In this appendix we show that, within Swirl–String Theory (SST), the modular parameter τ admits a natural geometric interpretation as an emergent clock–phase variable associated with compact relational time.

E.2 The Modular Parameter as a Relational Time Phase

In SST, physical time is defined operationally by a preferred foliation represented by a unit timelike vector field u^μ. Closed swirl–string configurations admit an internal phase associated with evolution along this foliation, encoded by the compact swirl–clock variable

$$S^{\circlearrowleft}_{(t)} \in \mathbb{S}^1. \tag{47}$$

We identify the modular parameter τ used in flavor models with the complex representation of this internal phase structure,

$$\boxed{\tau \;\longleftrightarrow\; \theta_{\text{geom}} + i\,\theta_{\text{stab}}} \tag{48}$$

where:

- θ_{geom} corresponds to a geometric winding or relative embedding phase of the swirl–string,
- θ_{stab} encodes stability and normalizability constraints imposed by finite swirl energy density.

In this interpretation, τ is not an arbitrary spurion but a derived, dynamical quantity determined by the geometry and energy content of the configuration.

E.3 Origin of Modular Functions

Closed, periodic configurations evolving along a compact clock parameter naturally admit functions that are invariant under discrete reparametrizations of the phase. The appropriate functional space is therefore that of modular forms.

Thus, modular invariance arises in SST as a kinematic consequence of:

- compact relational time,
- closed topological support of excitations,

- phase–relabeling invariance of the internal clock.

Modular forms do not represent additional symmetries imposed on the theory, but rather the natural basis functions on the configuration space of compact swirl–clock phases.

E.4 Mass Hierarchies and Modular Weights

In modular flavor models, fermion masses scale with modular forms of specific weights k. Within SST, the mass of a configuration K is determined by the integrated swirl energy density,

$$M_K = \mathcal{C} \int_{V_K} \rho_E(\mathbf{x})\, dV. \tag{49}$$

Discrete modular weights correspond to topological suppression factors arising from the distribution of swirl energy over increasing effective volumes. Thus,

$$k \longleftrightarrow n_K, \tag{50}$$

where n_K is the topological complexity index of the configuration. Mass hierarchies emerge as a consequence of energy dilution over larger or more intricate geometric support, rather than from arbitrary coupling matrices.

E.5 Mixing and CP Violation

In modular symmetry models, CP violation arises from the complex phase of τ. In SST, this phase has a direct geometric interpretation as the relative orientation of swirl–clock phases between different configurations.

Mixing matrices therefore arise from geometric misalignment of closed swirl–strings embedded in the same foliation, while CP–violating phases encode chirality and orientation information of the underlying topology. No fundamental complex Yukawa couplings are required.

E.6 Conceptual Consequences

This reinterpretation yields several conceptual clarifications:

- modular symmetry is emergent rather than fundamental,
- the parameter τ acquires a physical, geometric meaning,
- flavor structure, CP violation, and mass hierarchies share a common origin in relational time geometry,
- modular flavor models are recovered as effective infrared descriptions of SST.

E.7 Connection to Topological Mass Hierarchies

The interpretation of modular symmetry presented here connects directly to the topological mass hierarchy discussed in Appendix B. In that appendix, fermion masses arise from the integrated swirl energy density distributed over closed configurations with increasing topological complexity.

Within modular flavor models, mass suppression is encoded through modular forms of increasing weight k. In the SST framework, these modular weights correspond precisely to the topological complexity index n_K that controls the geometric suppression factor

$$\epsilon_K = \frac{r_c}{L_K}, \tag{51}$$

introduced in Appendix B.

The mapping

$$k \longleftrightarrow n_K \tag{52}$$

identifies modular weight as the infrared shadow of topological energy dilution in closed swirl–string configurations. What appears in modular symmetry constructions as an abstract algebraic grading is, in SST, a direct measure of how swirl energy is distributed over an increasingly extended or intricate geometric support.

Consequently, modular flavor models recover the same hierarchical mass patterns as Appendix B, but without access to the underlying geometric mechanism. Swirl–String Theory therefore unifies discrete modular hierarchies and topological mass suppression as two descriptions of the same physical origin.

E.8 Summary

Modular symmetry in flavor physics can be understood within SST as the infrared shadow of compact relational time and closed swirl–string topology. What appear as abstract modular parameters in phenomenological models correspond in SST to concrete geometric and energetic features of physical configurations.

F Parameter–Independent Mass Relations from SST Sector Ansatz

F.1 Motivation

In conventional flavor models, fermion masses are determined by Yukawa matrices containing many free parameters. Nevertheless, it has been shown that if a fermion mass matrix depends on a small number of underlying parameters, then algebraic relations between the physical masses necessarily follow [3].

Swirl–String Theory (SST) is particularly well suited for such relations, since fermion masses arise from geometric and energetic properties of closed configurations rather than from arbitrary couplings. In this appendix we show that SST generically predicts *parameter–independent mass relations* whenever a fermion sector is governed by a restricted set of topological or geometric degrees of freedom.

F.2 Invariant Structure of Mass Matrices

Consider a fermion sector f described effectively by a Hermitian matrix H_f, whose eigenvalues are proportional to the squared physical masses $m_{f1}^2, m_{f2}^2, m_{f3}^2$. Independently of the microscopic origin of H_f, its spectrum is fully characterized by the three invariants

$$I_1 = \mathrm{Tr}(H_f) = m_{f1}^2 + m_{f2}^2 + m_{f3}^2, \tag{53}$$

$$I_2 = \tfrac{1}{2}\left[\mathrm{Tr}(H_f)^2 - \mathrm{Tr}(H_f^2)\right] = m_{f1}^2 m_{f2}^2 + m_{f2}^2 m_{f3}^2 + m_{f3}^2 m_{f1}^2, \tag{54}$$

$$I_3 = \det(H_f) = m_{f1}^2 m_{f2}^2 m_{f3}^2. \tag{55}$$

These invariants are independent of basis choice and represent observable combinations of the fermion masses.

F.3 Elimination of Sector Parameters

Suppose that the effective SST description of sector f yields a mass matrix $H_f(\lambda_1, \ldots, \lambda_N)$ depending on N continuous parameters λ_i, where $N < 3$. Then the three invariants I_1, I_2, I_3 are functions of the same N parameters,

$$I_k = I_k(\lambda_1, \ldots, \lambda_N), \qquad k = 1, 2, 3. \tag{56}$$

Because the number of invariants exceeds the number of parameters, it is possible to eliminate the λ_i and obtain at least $(3-N)$ independent algebraic relations among the physical masses,

$$F_j(m_{f1}^2, m_{f2}^2, m_{f3}^2) = 0, \qquad j = 1, \ldots, 3-N. \tag{57}$$

These relations are *parameter independent* and therefore constitute genuine predictions of the theory.

F.4 Interpretation within Swirl–String Theory

In SST, the mass of a fermionic excitation is determined by the integrated swirl energy density over a closed configuration,

$$M_K = \mathcal{C} \int_{V_K} \rho_E(\mathbf{x})\, dV, \tag{58}$$

possibly modified by discrete topological suppression factors discussed in Appendix B.

If all configurations in a given fermion sector share the same underlying topological class, differing only by a small number of geometric or phase parameters, then the effective description of that sector necessarily involves few continuous degrees of freedom. Parameter–independent mass relations are therefore not accidental but an expected consequence of the SST ontology.

F.5 Relation to Quark Mass Relations

An explicit example of this mechanism is provided by the $U(3) \times U(3)'$ model of Koide and Nishiura, where quark mass matrices depend on a single complex parameter and yield a cubic mass relation independent of that parameter [3].

Within SST, such constructions are reinterpreted as effective infrared descriptions of a deeper geometric restriction: the quark sector is assumed to occupy a narrow class of admissible swirl–string configurations. The resulting mass relations reflect this geometric constraint rather than a specific flavor symmetry.

F.6 Predictive Power and Falsifiability

Parameter–independent mass relations provide a sharp experimental test of SST. Improved determinations of light quark masses or leptonic mass ratios can directly confirm or falsify the existence of the corresponding invariant constraints.

Moreover, different topological sectors are expected to yield different functional forms of F_j, allowing SST to be progressively constrained or ruled out by precision mass measurements.

F.7 Summary

Whenever a fermion sector in Swirl–String Theory is governed by fewer continuous degrees of freedom than observable mass parameters, algebraic relations among the masses necessarily emerge. Such parameter–independent mass relations are a robust and testable signature of the SST framework and unify previously phenomenological constructions within a single geometric interpretation.

G Exchange Statistics as Swirl–Clock Holonomy

G.1 Motivation

In standard quantum field theory, particle exchange statistics are postulated as either bosonic or fermionic, with the spin–statistics theorem providing a consistency condition under locality and relativistic causality. Recent work has demonstrated, however, that exchange statistics are

more generally determined by the topology of configuration space and the internal structure of excitations, rather than by point-particle assumptions alone [1].

Swirl–String Theory (SST) naturally accommodates this broader perspective. Its fundamental excitations are extended, closed configurations evolving along a preferred foliation, endowed with a compact internal clock phase. This appendix shows that, within SST, exchange statistics arise as a geometric holonomy of the swirl–clock phase under nontrivial exchange paths in configuration space.

G.2 Exchange as a Configuration-Space Loop

Consider two identical SST excitations described by closed configurations K_1 and K_2. An exchange process is not merely a permutation of labels but a continuous deformation of the joint configuration through configuration space. Such a process defines a closed loop γ in the reduced configuration space modulo gauge and foliation redundancies.

The physically relevant effect of this loop is the accumulated phase of the internal clock variable along γ,

$$\Delta\theta = \oint_\gamma dS^{\circlearrowleft}_{(t)}. \tag{59}$$

This phase is invariant under smooth deformations of the exchange path that do not cross singular configurations and therefore represents a topological holonomy.

G.3 Classification of Statistics

The exchange operator $\mathcal{E}$ acting on the joint state of two identical excitations acquires the form

$$\mathcal{E} = \exp(i\,\Delta\theta)\,, \tag{60}$$

where $\Delta\theta$ is determined by the topology of the exchange path and the internal geometry of the configurations.

Special cases recover familiar statistics:

- $\Delta\theta = 0 \pmod{2\pi}$ corresponds to bosonic exchange,
- $\Delta\theta = \pi \pmod{2\pi}$ corresponds to fermionic exchange,
- intermediate or matrix-valued holonomies correspond to generalized or composite statistics.

Thus, exchange statistics in SST are not postulated but emerge from the geometry of configuration space and the compactness of relational time.

G.4 Relation to Spin–Statistics

The standard spin–statistics connection is recovered in SST as a consistency condition imposed by relativistic causality and foliation-compatible locality in the infrared. Elementary excitations with minimal topological structure admit only trivial or π holonomy, reproducing bosons and fermions.

More intricate closed configurations may admit richer holonomy classes without violating unitarity or causality, provided the exchange paths remain compatible with the foliation structure. In this sense, SST generalizes rather than contradicts the conventional spin–statistics connection.

G.5 Time, Memory, and Nonlocality

A key implication of the holonomy interpretation is that exchange statistics depend on the history of the configuration rather than solely on instantaneous positions. This reflects the presence of internal memory encoded in the swirl–clock phase.

Such behavior is compatible with quantum mechanics because the memory is topological rather than dynamical: it depends only on the homotopy class of the exchange path. No violation of microcausality or signal locality is implied.

G.6 Conceptual Consequences

The holonomy interpretation of exchange statistics yields several conceptual clarifications:

- statistics are emergent properties of extended configurations,
- bosonic and fermionic behavior are limiting cases of a broader geometric classification,
- quantum phase, time evolution, and exchange are unified through the swirl–clock variable,
- no modification of quantum mechanics or Lorentz symmetry is required in the infrared.

G.7 Summary

In Swirl–String Theory, particle exchange statistics arise from the holonomy of the compact swirl–clock phase associated with nontrivial loops in configuration space. This framework reproduces conventional bosonic and fermionic statistics while allowing a principled extension to more general topological classes for composite or highly structured excitations. Exchange statistics are thus understood as a manifestation of relational time geometry rather than as an independent axiom.

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